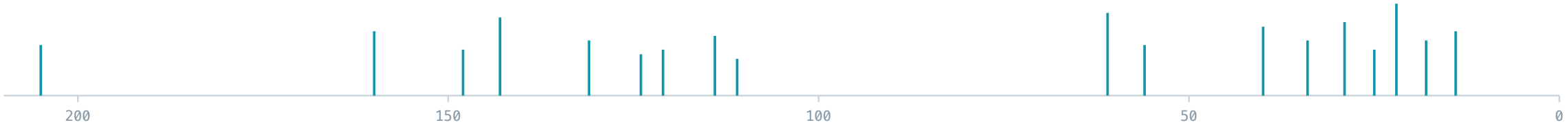
COMPUTER-ASSISTED STRUCTURE ELUCIDATION · FROM NMR

AILSA

AI + LSD + Agents

An AI agent that aims to autonomously determine the structure of an unknown organic compound from its NMR spectra — reasoning about the data, building solver constraints, and ranking candidate structures without human intervention.



Domain skill-set
extensive NMR / CASE expertise & workflow strategy — the intelligence layer

+

Thin tools
nmrglue · LSD · RDKit CLI wrappers for data & solvers

+

Agent team
Claude Code's agent-team feature — specialists that coordinate with each other

= one autonomous CASE system, orchestrated entirely inside **Claude Code**.

github.com/steinbeck/ailsa

Spectra in, structure out — reasoned, not guessed

A chemist elucidating a natural product juggles ^{1}H, ^{13}C, HSQC, HMBC and COSY, forms hypotheses, and rejects the ones the data contradicts. AILSA puts that reasoning loop under an AI agent.

The Python tools stay deliberately thin — wrappers around **nmrglue**, **LSD** and **RDKit**. Domain expertise and strategy live in the agent skill, not in hard-coded heuristics.

The hard part

Under-determined by design

NMR gives connectivity fragments, not a structure. Many candidates fit the same data — the agent must generate, rank, and rule out.

The guardrail

A team, not a lone agent

A multi-agent architecture prevents unproductive loops and the "clean-but-wrong" answer: low error, plausible, yet the wrong isomer.

The solver

Emergent, not forced

Rings and skeletons arise from MULT + HMBC + COSY constraints — not from a hand-placed answer. Verified: a benzene ring emerged with zero ring bonds asserted.

THE PIPELINE BRUKER DATA → ANSWER

From raw Bruker spectra to a ranked structure

01 · READ

NMR data

1D ^{1}H / ^{13}C and 2D HSQC · HMBC · COSY read straight from Bruker files.

`nmrglue`

02 · PICK

Peak picking

Raw picking with SNR floors so weak quaternary carbons survive; the chemist agent applies DEPT/HMBC guidance.

`ailsa pick`

03 · DETECT

Statistical detection

Hybridisation, forbidden/mandatory neighbours & hetero-hetero bonds from 7.9M HOSE stats.

`ailsa detect`

04 · BUILD

Constraints + fragments

MULT / HSQC / HMBC / BOND plus DEFF fragment goodlists, written into the LSD file by the LSD-engineer agent.

`lsd-engineer agent`

05 · SOLVE

LSD structure generator

Nuzillard's deterministic solver enumerates every structure consistent with the constraints.

`LSD (Reims)`

06 · RANK

^{13}C prediction

HOSE-code shift prediction scores candidates; match-count first, MAE second — no hallucinated fits.

`ailsa lsd rank`

07 · CHECK

Identity gate

Derive identity from the SMILES via **ailsa identify** — never a recalled name; RDKit-verified.

`ailsa identify`

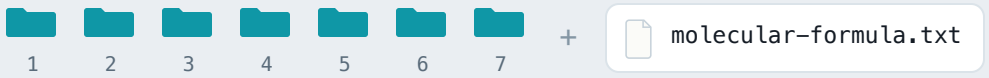
DEREPLICATE

Known compound?

Optional formula-indexed lookup against 928K compounds runs as a separate path — never inside CASE.

`ailsa dereplicate`

Input — one data directory per unknown compound



Numbered **Bruker experiment folders** (^{1}H · ^{13}C · HSQC · HMBC · COSY) · plus the known molecular formula

Four specialists, one coordinator

case.md · coordinator

Starts the team, writes the single source-of-truth **CASE-PROGRESS.md**, detects unproductive loops, and intervenes with targeted advisories. Escalates to **ailsa-diagnostic** for deep LSD failure analysis.

spectra

ailsa-nmr-chemist

Peak picking, statistical detection, spectral-quality assessment, and the chemistry-first evidence hierarchy — NMR always overrides statistics.

solver

ailsa-lsd-engineer

Builds LSD constraint files, iterates HMBC incrementally, owns the constraint inventory (read previous, never reconstruct), runs the solver.

ranking

ailsa-solution-analyst

Ranks solutions by ^{13}C prediction, judges chemical plausibility, scores confidence, and writes the final result with a tentative identity.

quality gate

ailsa-devils-advocate

Validates every LSD file before it runs, checks constraint persistence across iterations, and enforces the G-IDENT / G-MULT gates against wrong-but-clean answers.

ailsa-lsd-engineer ⇄ ailsa-devils-advocate · direct review loop — every LSD file is reviewed before the solver runs

ailsa-lsd-engineer → ailsa-nmr-chemist · requests — ask the NMR chemist for more correlations or detection results

Push-never-pull coordination. The coordinator sends each specialist a `[BEGIN]` after creating its task; agents never poll. From there the specialists that coordinate with each other work through the direct review loop and ask the NMR chemist when they need more data; the coordinator stays the **sole writer** of the progress log.

DATA SOURCE NMRXIV.ORG

Open, FAIR spectra — fetched by DOI

nmrXiv is an open, community repository for raw and processed NMR data. Every dataset used here — the worked examples and all 256 benchmark datasets — is public, citable, and pulled straight from it — no proprietary files, no vendor lock-in.

FETCH A DATASET

```
$ ailsa fetch nmrxiv 10.57992/NMRXIV.P9.S51
→ downloads the Bruker experiment folders into ./
```

HOW IT IS ORGANISED

Project *Pn* → **Study** *Sn* → **Dataset** · Bruker

Why it matters — openly citable benchmarks let anyone re-run a CASE evaluation on the exact same experimental data.

WHERE THE CASE EXAMPLES COME FROM

P9	**Teaching Datasets 1**	Duisburg-Essen · CASE1, CASE9
P42	**Classics in Spectroscopy**	CASE2, CASE4, CASE5, CASE8
P124	**Alkaloids of Virgilia Oroboides**	Krytovych / Breuning · CASE7

CASE3 & CASE6 come from further public nmrXiv datasets (pulegone, citronellol).

BLIND BENCHMARK 256 PUBLIC DATASETS NMRXIV

Blind on 256 public datasets: 69.1 % solved at rank 1

Measured in a restricted configuration: without the agent team (unattended runs form none, so the specialists ran one at a time under the coordinator) and without statistical detection (outdated reference database on the compute host).

HOW IT IS MEASURED

01

Spectra in, nothing else

The coordinator gets a sanitised directory of Bruker spectra and the molecular formula, never a name or structure; dereplication is forbidden.

02

Answers locked away

Every file mapping a dataset to its answer is moved out of reach for the duration of each batch; a system-prompt fence forbids looking for it.

03

Graded from outside

An independent grader re-derives every candidate's InChIKey with RDKit; the run's own claim is never used.

04

Constitution only

Stereochemistry is not tested.

177 / 256
correct structure at rank 1
69.1 %

189 / 256
within the top 10
73.8 %

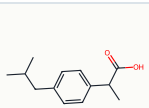
≤ 15 heavy atoms	40 / 40	100 %
16–20	28 / 32	88 %
21–25	49 / 70	70 %
≥ 26	60 / 114	53 %

Same 102 datasets, two model generations: Opus 5 67 / 102 vs Opus 4.8 40 / 102 at rank 1 — won 30, lost 3, exact McNemar $p = 1.4 \cdot 10^{-6}$

The limit is size. Up to 15 heavy atoms every molecule was solved; from 26 upward about half. When a run fails, the right structure is almost never further down its list — generation, not ranking, is what limits it. 25 runs hit the three-hour limit and count as failures.

Nine worked examples

Illustrative public datasets, separate from the benchmark. Ground truth derived independently (nmrXiv + ^{13}C verification); each run graded by a fresh blind instance, RDKit-verified.

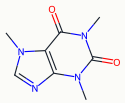


CASE1 Solved

Ibuprofen

$C_{13}H_{18}O_2$ nmrxiv P9 / S51

Benzene ring emerged from constraints — 0 ring-BONDs; Rank 1, exact InChIKey

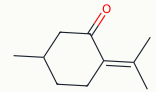


CASE2 Solved

Caffeine

$C_8H_{10}N_4O_2$ nmrxiv P42 / S1328

Purine alkaloid; two C=O + three N-CH_3 pin the xanthine core

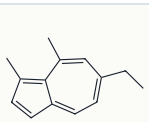


CASE3 Solved

(R)-Pulegone

$C_{10}H_{16}O$ nmrxiv pulegone set

α,β-unsat. ketone; ^{13}C rules out the formula-twin thujone

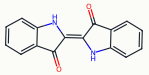


CASE4 Partial

Chamazulene

$C_{14}H_{16}$ nmrxiv P42 / S1312

Correct azulene scaffold — exact substitution regiochemistry not yet reachable

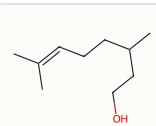


CASE5 Solved

Indigo

$C_{16}H_{10}N_2O_2$ nmrxiv P42 / S1302

C_2-symmetric; single C=O rules out asymmetric indirubin

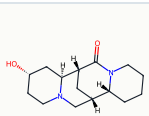


CASE6 Solved

(R)-Citronellol

$C_{10}H_{20}O$ nmrxiv citronellol set

Acyclic monoterpenoid; olefinic pair + CH_2OH rule out menthol

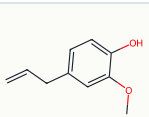


CASE7 Solved

Virgiline

$C_{15}H_{24}N_2O_2$ nmrxiv P124 / S1266

Tetracyclic lupin alkaloid; independently re-validated on a clean host

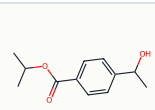


CASE8 Solved

Eugenol

$C_{10}H_{12}O_2$ nmrxiv P42

Allyl-phenol; terminal =CH_2 rules out isoeugenol



CASE9 Solved

Isopropyl 4-(1-hydroxyethyl)benzoate

$C_{12}H_{16}O_3$ nmrxiv P9 / S53 (D202)

First-ever full solve (2026-06-11); benzoate ester, MAE 1.17

BY THE NUMBERS SCALE & PROVENANCE

What the agent stands on

928K
reference compounds
COCONUT (predicted) + NMRShiftDB (experimental)

7.9M
HOSE statistics
^{13}C shift prediction

2.4M
fragment SSCs
DEFF goodlist search

111K
unique formulas
formula-indexed lookup

1372
passing tests
pytest · mypy-strict · ruff

69.1%
rank 1 in the blind benchmark
177 / 256 · restricted configuration, see slide 6

16
CLI command groups
--format json on all analysis commands

5
agents in the team
coordinator + 4 specialists

BUILT ON

LSD — the deterministic structure generator by Jean-Marc Nuzillard (Université de Reims). **HOSE codes** — Bremser, 1978. Parsing via **nmrglue**; cheminformatics via **RDKit**. **Reference data** — COCONUT and NMRShiftDB; COCONUT 13C shifts: from J.-M. Nuzillard, Zenodo, doi:10.5281/zenodo.6561449, CC BY 4.0.

GET IT

github.com/steinbeck/ailsa · **pip install ailsa** · reference database DOI **10.6084/m9.figshare.31073554**